Microneedle Superdisintegrant Film Simulator v5
Comprehensive Physics, Mathematics, Kinetics, and Software Architecture

Technical Documentation

Contents

1 Purpose of the Simulator

This simulator predicts the behavior of coated microneedles containing drug-loaded HPMC films with superdisintegrants. It combines:

- Microneedle geometry
- TPMS porous architectures
- Polymer hydration
- Capillary wicking
- Drug release kinetics
- Formulation effects
- Drug loading calculations

The model is a hybrid semi-mechanistic framework. Some portions originate from classical transport theory, while others are empirical calibration relationships chosen to reproduce expected formulation trends.

2 Simulation Workflow

1. User specifies geometry.
2. User specifies formulation.
3. Material properties are calculated.
4. Geometric properties are calculated.
5. Drug loading is calculated.
6. Wicking profile is calculated.
7. Hydration kinetics are calculated.
8. Burst release kinetics are calculated.
9. Sustained release kinetics are calculated.
10. Total release profile is generated.
11. Metrics such as T50 and T80 are extracted.

3 Material Model

3.1 Inputs

Drug MW	Molecular weight (Da)
HPMC grade	E5, E15, E50, K4M
HPMC concentration	wt%
Tween 80 concentration	wt/v%
Glycerol concentration	wt%

Superdisintegrant type	CCS, SSG, PVPP
Superdisintegrant concentration	wt%

3.2 Effective Water Diffusivity

Implemented in:

`effectiveDwater()`

The simulator assumes

$$D_w = 10^{-10} \left(\frac{5}{\mu_{ref}} \right)^{0.28} \exp(-0.03 C_{HPMC}) \tag{1}$$

where

- D_w = effective water diffusivity
- μ_{ref} = viscosity reference for HPMC grade
- C_{HPMC} = HPMC concentration

Physical interpretation:

- Higher molecular entanglement reduces water transport.
- Increasing polymer concentration reduces free volume.
- K4M creates the strongest diffusion barrier.

4 Viscosity Model

The simulator estimates coating viscosity using

$$\mu = \mu_{ref} \left(\frac{C_{HPMC}}{2} \right)^{2.2} \tag{2}$$

Examples:

Grade	Conc (%)	Estimated Viscosity
E5	5	$\approx$ 37 mPa s
E15	5	$\approx$ 112 mPa s
E50	5	$\approx$ 375 mPa s
K4M	5	$\approx$ 30000 mPa s

5 Surface Tension Model

Tween-80 lowers surface tension according to

$$\gamma = 72 - 35 \log_{10} \left(1 + \frac{C_T}{0.01} \right) \tag{3}$$

bounded by

$$28 \le \gamma \le 72 \tag{4}$$

mN/m.

6 Superdisintegrant Physics

6.1 CCS

Mechanism:

- Swelling
- Wicking
- Fiber expansion

Default parameters:

$$k_0 = 0.14 \tag{5}$$

$$n_0 = 0.55 \tag{6}$$

6.2 SSG

Mechanism:

- Rapid swelling
- Gel formation at high concentration

Default:

$$k_0 = 0.11 \tag{7}$$

$$n_0 = 0.65 \tag{8}$$

6.3 PVPP

Mechanism:

- Capillary action
- Minimal gel formation

Default:

$$k_0 = 0.23 \tag{9}$$

$$n_0 = 0.40 \tag{10}$$

7 Gel Penalty Model

Above a critical concentration:

$$P_{gel} = 0.75^{(C-C_{crit})} \tag{11}$$

Consequences:

- Reduced hydration
- Reduced burst release
- Reduced overall release rate

8 Microneedle Geometry

8.1 Aspect Ratio

$$AR = \frac{H}{R} \tag{12}$$

8.2 Tip Angle

$$\theta = \tan^{-1}\left(\frac{R}{H}\right) \tag{13}$$

9 Surface Area Models

9.1 Planar Film

$$SA = \pi R^2 \tag{14}$$

9.2 Conical Needle

$$s = \sqrt{H^2 + (R - r_t)^2} \tag{15}$$

$$SA = \pi(R + r_t)s \tag{16}$$

9.3 Pyramidal Needle

$$SA = 4\left(\frac{1}{2}as\right) \tag{17}$$

9.4 Star Needle

The simulator applies a 1.45 surface enhancement multiplier.

9.5 TPMS Needle

$$SA_{eff} = SA_{external} \times SA_{factor} \tag{18}$$

This approximates internal pore surface.

10 Drug Loading Calculations

Coating volume:

$$V_{coat} = SA \cdot L \cdot f_v \tag{19}$$

Film density:

$$\rho = 1.2\ g/cm^3 \tag{20}$$

Coating mass:

$$M_{coat} = V_{coat}\rho \tag{21}$$

Drug mass:

$$M_{drug} = f_{drug} M_{coat} \tag{22}$$

11 TPMS Transport Physics

Effective diffusivity ratio:

$$\frac{D_{eff}}{D} = \frac{\varepsilon}{\tau^2} \tag{23}$$

where

- ε = porosity
- τ = tortuosity

Used for:

- Gyroid
- Primitive
- Diamond
- Hybrid TPMS-core cone

12 Lucas-Washburn Wicking

The simulator solves

$$x^2 = Ct \tag{24}$$

or

$$x = \sqrt{Ct} \tag{25}$$

where

$$C = \frac{r\gamma \cos\theta}{2\eta\tau^2} \tag{26}$$

Physical meaning:

- Larger pores increase penetration.
- Lower viscosity increases penetration.
- Higher surface tension increases penetration.
- Larger tortuosity decreases penetration.

13 Hydration Physics

Hydration constant:

$$k_{hyd} = \frac{D_w}{L^2} \times 60 \tag{27}$$

Film hydration:

$$H(t) = 1 - e^{-k_{hyd}t} \tag{28}$$

Hydration reaches:

$$63.2\% \tag{29}$$

at

$$t = \frac{1}{k_{hyd}} \tag{30}$$

14 Release Kinetics

14.1 Korsmeyer-Peppas

$$F = k_{KP} t^n \tag{31}$$

where

- F = fractional release
- k_{KP} = release coefficient
- n = release exponent

14.2 Mechanism Classification

n	Mechanism
≤ 0.45	Fickian
0.45–0.89	Anomalous
≥ 0.89	Case II

15 Burst Release

Burst release fraction:

$$F_b = B\left(1 - e^{-k_b t}\right) \tag{32}$$

where

$$k_b = \frac{k_{hyd} k_{burst}}{k_{hyd} + k_{burst}} \tag{33}$$

This is effectively a harmonic combination of hydration and burst kinetics.

16 Combined Release Model

Total release:

$$F_{total} = F_{burst} + (1 - B) F_{sustained} \tag{34}$$

where

$$0 \leq F_{total} \leq 1 \tag{35}$$

17 Kinetic Profile Interpretation

17.1 T50

Time for

$$F = 0.5 \tag{36}$$

17.2 T80

Time for

$$F = 0.8 \tag{37}$$

Fast-release systems typically achieve

$$T_{80} < 10\ min \tag{38}$$

while sustained systems exceed 20 min.

18 TPMS Mathematical Definitions

18.1 Gyroid

$$\sin x \cos y + \sin y \cos z + \sin z \cos x = 0 \tag{39}$$

18.2 Primitive

$$\cos x + \cos y + \cos z = 0 \tag{40}$$

18.3 Diamond

$$\sin x \sin y \sin z + \sin x \cos y \cos z + \cos x \sin y \cos z + \cos x \cos y \sin z = 0 \tag{41}$$

19 Code Architecture

19.1 materialModel()

Computes:

- viscosity
- surface tension
- diffusivity
- wetting factor
- burst factor
- gel penalty

19.2 geometryModel()

Computes:

- area
- volume
- drug load
- TPMS corrections

19.3 simulateWicking()

Computes Lucas-Washburn penetration.

19.4 simulateRelease()

Computes:

- hydration
- burst release
- sustained release
- total release

19.5 buildSim()

Master orchestration function.

20 Parameter Sensitivity

Largest positive influence on release:

1. Higher porosity
2. Lower thickness
3. Higher CCS/PVPP content
4. Higher Tween concentration
5. Lower HPMC concentration

Largest negative influence:

1. K4M
2. High HPMC loading
3. Excessive SSG
4. Large tortuosity
5. Thick coatings

21 Literature Basis

Likely scientific foundations:

1. Korsmeyer RW et al., controlled release systems.
2. Peppas NA, polymeric drug delivery.
3. Lucas and Washburn capillary penetration theory.
4. Millington and Quirk effective diffusivity.
5. HPMC oral film literature.
6. Dissolving microneedle drug delivery literature.
7. TPMS transport and porous media theory.
8. CCS, SSG and PVPP superdisintegrant studies.

22 Limitations

The simulator does not include:

- Finite-element diffusion
- Skin diffusion resistance
- Drug crystallization
- Polymer erosion mechanics
- Concentration-dependent diffusivity
- Coupled swelling-deformation physics
- Mechanical fracture of microneedles

Therefore outputs should be interpreted as engineering estimates rather than absolute predictions.